9th Class • Punjab Board • PCTB New Syllabus

Chapter 11: Biostatistics

Solved notes in book order — MCQs, short questions, long questions, important questions and key points.

Chapter 11 • Units 11.1 – 11.3 • Exercise solved in book order

What this chapter is about

• Biostatistics applies statistical methods to biological data. It helps scientists design experiments, analyse results, predict outcomes and make evidence-based decisions.
• Mean, median and mode summarise a data set in different ways. A bar chart compares the values of different categories visually.
• The solved exercise includes complete numerical working for mean, median and mode, step-by-step Excel instructions and the required fruit-sales bar chart.
• The textbook prints no Inquisitive Questions section in this chapter, so none has been invented.

A. Multiple Choice Questions

Select the correct option. Calculations and reasons are included where useful.

1. What is the primary purpose of biostatistics?

a) To analyse financial data
b) To apply statistical methods to biological sciences
c) To design engineering models
d) To study historical events

Answer: b) To apply statistical methods to biological sciences — Biostatistics uses numerical methods to design, analyse and interpret biological studies.

2. In biostatistics, which method is used to predict future outcomes based on current data?

a) Designing experiments
b) Drawing charts
c) Taking average
d) Analysing data

Answer: d) Analysing data — Patterns and relationships found by analysing current data can be used to build models and predict future outcomes.

3. Which of the following best describes the mean of a data set?

a) The most frequently occurring value
b) The middle value when data is ordered
c) The sum of all values divided by the number of values
d) The difference between the highest and lowest values

Answer: c) The sum of all values divided by the number of values — This is the arithmetic mean, or average, of the data set.

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4. If the data set is 5, 8, 12, 15, 20, what is the median?

a) 8
b) 12
c) 15
d) 20

Answer: b) 12 — The five values are already ordered, so the third (middle) value is the median: 12.

5. What is the mean of the data set: 7, 8, 9, 10, 11?

a) 7
b) 8
c) 9
d) 10

Answer: c) 9 — Mean = (7 + 8 + 9 + 10 + 11) ÷ 5 = 45 ÷ 5 = 9.

6. When the number of values in a data set is even, how is the median calculated?

a) By choosing the middle value
b) By taking the average of the two middle values
c) By selecting the most frequent value
d) By adding all values and dividing by the total number of values

Answer: b) By taking the average of the two middle values — After ordering the data, add the two central values and divide their sum by two.

7. In a data set with values 3, 3, 6, 7, 8, 9, 9, what is the mode?

a) 3
b) 6
c) 7
d) Both 3 and 9

Answer: d) Both 3 and 9 — Both 3 and 9 occur twice, which is the highest frequency, so the data set is bimodal.

8. If a data set has no repeated values, what is the mode?

a) The highest value
b) The average of the data set
c) There is no mode
d) The median value

Answer: c) There is no mode — A mode must have the highest repeated frequency; if every value occurs once, no mode exists.

9. In a bar chart, what does the height or length of each bar represent?

a) The total number of categories
b) The value of the corresponding category
c) The average of all values
d) The difference between highest and lowest values

Answer: b) The value of the corresponding category — A taller or longer bar shows a greater numerical value or frequency for that category.

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10. When constructing a bar chart, which axis usually represents the categories?

a) Vertical axis (y-axis)
b) Horizontal axis (x-axis)
c) Both axes equally represent the categories
d) Neither axis represents the categories

Answer: b) Horizontal axis (x-axis) — In a vertical bar chart, category names are placed on the x-axis and their numerical values on the y-axis.

B. Write Short Answers

1. Define biostatistics.

Biostatistics is the branch of statistics that applies statistical methods to the biological sciences. It is used to design experiments and clinical trials, collect and analyse biological data, interpret results and predict outcomes.

2. What is the median of a data set?

The median is the middle value after the data have been arranged in ascending or descending order. For an odd number of values, it is the central value. For an even number, it is the average of the two central values.

3. How is the mean calculated?

To calculate the arithmetic mean, add all the values and divide their sum by the total number of values.

$$\text{Mean} = \text{sum of all data points} \div \text{number of data points}$$

For example, the mean of 4, 6 and 8 is $(4 + 6 + 8) \div 3 = 18 \div 3 = \mathbf{6}$.

4. What does the height of a bar in a bar chart represent?

In a vertical bar chart, the height of a bar represents the numerical value or frequency of its category. A taller bar means a larger value. The same scale must be used for every bar so that the comparison remains fair.

5. What is the mode of a data set?

The mode is the value that occurs most frequently in a data set. A data set may have one mode, two modes (bimodal), more than two modes (multimodal), or no mode when all values occur with the same frequency.

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C. Write Answers in Detail

1. Explain the importance of biostatistics in the field of public health. Provide examples of how it is used to inform public health decisions.

Public health decisions affect whole populations, so they should be based on reliable evidence rather than guesses. Biostatistics helps health workers collect enough representative data, analyse them correctly and interpret whether a result is real or may be due to chance.

1. Measuring disease patterns: Data on infection, recovery and death rates show which diseases are common, which groups are at greater risk and where an outbreak is spreading. For example, analysis of COVID-19 infection rates helped authorities identify high-risk areas and plan control measures.

2. Predicting outbreaks: Current and earlier data can be used to estimate how quickly an infectious disease may spread. These predictions help hospitals prepare beds, staff, medicines and diagnostic facilities before the number of patients rises sharply.

3. Planning vaccination: Biostatistics is used to test vaccine safety and effectiveness in clinical trials and to compare infection rates in vaccinated and unvaccinated groups. It also helps estimate how many doses are needed and which areas or age groups should receive them first.

4. Evaluating health programmes: Statistics collected before and after a programme show whether it worked. For example, a fall in diarrhoea cases after a clean-water campaign supports continuing that campaign; no improvement shows that the plan needs changing.

5. Guiding policy and resources: Evidence about maternal health, nutrition, smoking, pollution and other risk factors helps governments choose useful policies and direct limited funds to the greatest needs. In this way, biostatistics turns population data into fair and evidence-based public health action.

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2. Discuss the differences between mean, median, and mode. Include examples where each measure is most appropriate to use.

Three measures of central tendency

Mean	Median	Mode
The arithmetic average of the data.	The middle value after ordering the data.	The most frequently occurring value.
Add all values and divide by the number of values.	For odd n, select the central value; for even n, average the two central values.	Count the frequency of each value and select the highest frequency.
Example: for 4, 6, 8, mean = 18 ÷ 3 = **6**.	Example: for 2, 4, 100, median = **4**.	Example: for 2, 3, 3, 5, mode = **3**.

Mean	Median	Mode
Uses every value and is useful for roughly balanced numerical data.	Best when data are skewed or contain extreme values, because it resists outliers.	Best for the most common measurement or for categorical data.
Example use: average plant height of similar plants in an experiment.	Example use: typical recovery time when one patient takes an unusually long time.	Example use: most common blood group or most frequent shoe size.
Limitation: extreme values can pull the mean up or down.	Limitation: it ignores the exact size of values away from the centre.	Limitation: there may be no mode or more than one mode.

The best measure therefore depends on the data and the question. Mean describes the overall average, median describes the central position, and mode identifies the most common observation.

3. Describe the steps involved in creating a bar chart using Excel. Include a discussion on how to customize the chart for better visualization and interpretation of data.

1. Enter and organise the data: Type category names in one column and their numerical values in the next column. Give both columns clear headings and check that no value is missing or entered in the wrong row.

2. Select the data: Highlight the category names, values and headings. Open the **Insert** tab, choose **Insert Column or Bar Chart**, and select **Clustered Column** for vertical bars or **Clustered Bar** for horizontal bars.

3. Add a meaningful title: Use a title that states what the data show, including the place or time if relevant, e.g. 'Fruits Sold at a Market in One Week'.

4. Label the axes: Through **Chart Elements → Axis Titles**, label the x-axis with the category and the y-axis with the measured value or unit. For example, 'Type of fruit' and 'Number of fruits sold'.

5. Adjust the scale: Format the value axis so that it normally starts at zero and uses equal intervals. A suitable maximum and interval make differences clear without exaggerating them.

6. Improve readability: Add data labels to show exact values, use a legible font, keep bars of equal width with even gaps, and use colours with good contrast. Avoid unnecessary 3-D effects and too many colours.

7. Use supporting elements carefully: Keep light gridlines if they help reading, and show a legend only when more than one data series is present. Change gap width or bar direction if category names are crowded.

8. Check and interpret: Compare every bar with the source table. The finished chart should make the largest, smallest and intermediate categories easy to identify

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and should allow a correct conclusion at a glance.

4. Provide a detailed example of how to calculate the mean, median, and mode of a data set. Use the following data set for your calculations: 12, 15, 22, 8, 19, 25, 15.

1. Mean

Add all seven values and divide the sum by 7:

Sum = 12 + 15 + 22 + 8 + 19 + 25 + 15

Sum = 116

Mean = 116 ÷ 7

Mean = 16.5714...

Mean ≈ 16.57

2. Median

First arrange the values in ascending order:

8, 12, 15, **15**, 19, 22, 25

Number of values = 7 (odd)

Middle position = (7 + 1) ÷ 2 = 4th value

Median = 15

3. Mode

The value **15** occurs twice; every other value occurs once. Therefore, 15 has the highest frequency.

Mode = 15

Final results

• Mean = **16.57** (approximately) • Median = **15** • Mode = **15**.

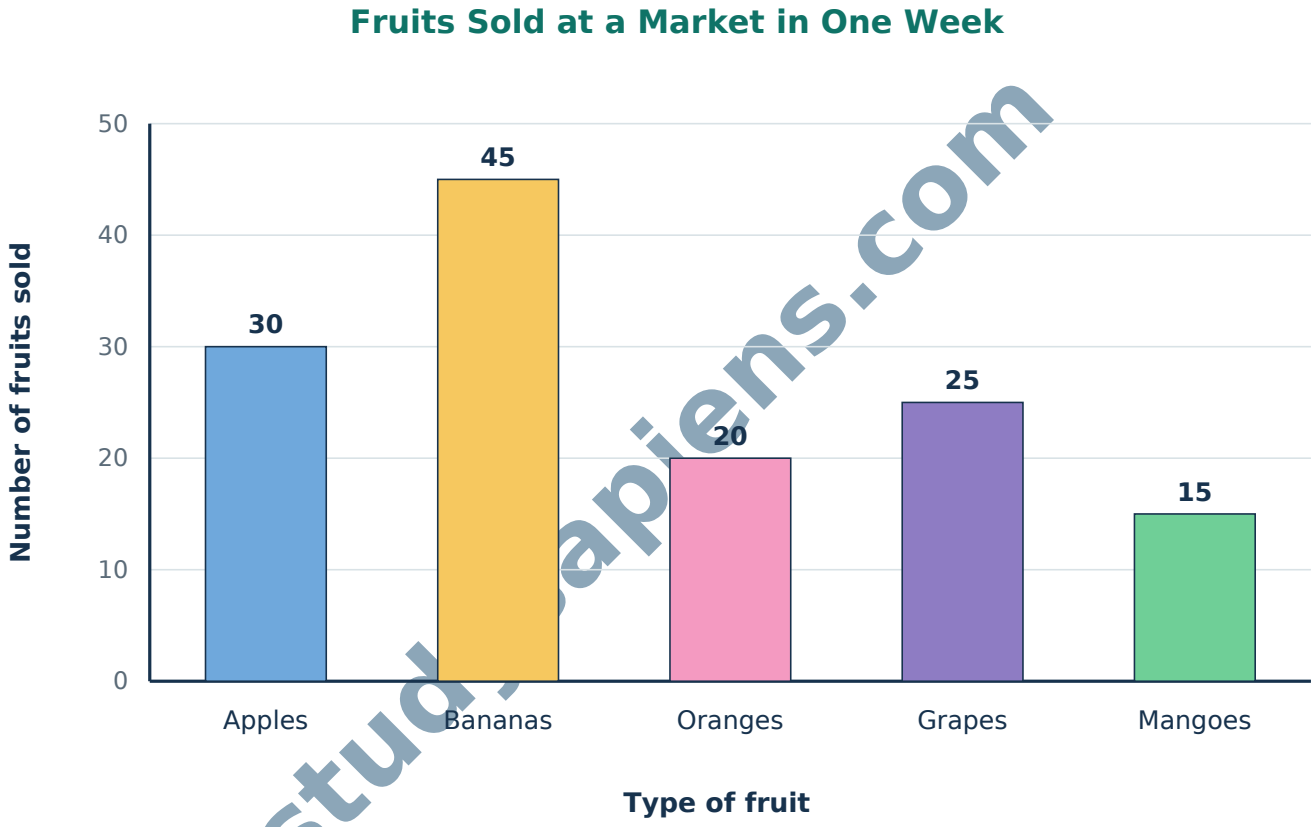
• The mean is slightly higher than the median because larger values such as 22 and 25 raise the average.

5. You are given the following data set, create a bar chart to represent the number of different types of fruits sold at a market in one week: Apples: 30; Bananas: 45; Oranges: 20; Grapes: 25; Mangoes: 15. Ensure that you label the axes and provide a title for the chart.

Data table

Type of fruit	Number sold in one week
Apples	30
Bananas	45
Oranges	20
Grapes	25
Mangoes	15

Required bar chart (y-axis scale: 1 division = 10 fruits)



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The chart shows that **bananas** had the highest sales (45), while **mangoes** had the lowest sales (15). Every bar has equal width and the bars are evenly separated.

D. More Important Questions (from the text)

1. State uses of biostatistics in genetics, agriculture and clinical trials.

Genetics: It analyses inheritance patterns and associations between genetic variations and diseases. For example, it helps researchers identify genetic risk factors linked with diabetes or cancer.

Agriculture: It compares crop yields, fertiliser effects and resistance to pests or diseases. For example, mean yields of several wheat varieties can be compared under the same field conditions.

Clinical trials: It helps decide sample size, assign suitable comparison groups and test whether a new treatment performs better than a placebo or an existing treatment. Correct analysis reduces the chance of accepting an ineffective drug.

2. What is meant by unimodal, bimodal, multimodal and no-mode data sets?

Type	Meaning and example
Unimodal	One mode: in 2, 3, 3, 5, the mode is 3.
Bimodal	Two modes with the same highest frequency: in 3, 3, 6, 9, 9, the modes are 3 and 9.
Multimodal	More than two modes share the highest frequency: in 2, 2, 4, 4, 7, 7, the modes are 2, 4 and 7.
No mode	All values have the same frequency: in 5, 8, 12, 15, each value occurs once.

3. Why can the median be more useful than the mean for skewed data?

The mean uses the exact size of every value, so one extremely large or small value can pull it away from most observations. The median depends only on the central position after sorting, so it is much less affected by that extreme value. For example, for 2, 3, 4, 5, 100, the mean is 22.8 but the median is 4; the median better represents the four ordinary values.

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Points to Remember

Biostatistics: application of statistical methods to biological sciences.

Main uses: designing experiments, analysing biological data, interpreting results, predicting outcomes and informing public health policy.

Fields: epidemiology, genetics, agriculture and clinical trials all depend on biostatistics.

Mean: sum of all values ÷ number of values. It uses every observation but is affected by extreme values.

Median: middle value of ordered data. For even n, take the average of the two middle values. It is useful for skewed data.

Mode: most frequent value. A data set may be unimodal, bimodal, multimodal or have no mode.

Bar chart: bars of different heights or lengths compare numerical values of different categories.

Axes: categories normally go on the x-axis and values on the y-axis in a vertical bar chart.

Good chart: clear title, labelled axes with units, zero-based equal scale, equal bar widths, even gaps and accurate data labels.

Before calculating: sort the data for median, count frequencies for mode and check the number of observations for mean.

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How to attempt this chapter in the paper

• Show every numerical step: formula → substitution → calculation → final answer, rounding only at the end when required.
• For median, never start before arranging the data in ascending or descending order.
• A bar chart is not complete without a title, both axis labels and a clear scale. Include exact values above the bars where possible.
• For public-health answers, give specific examples such as outbreak prediction, vaccination planning and evaluation of a health programme.